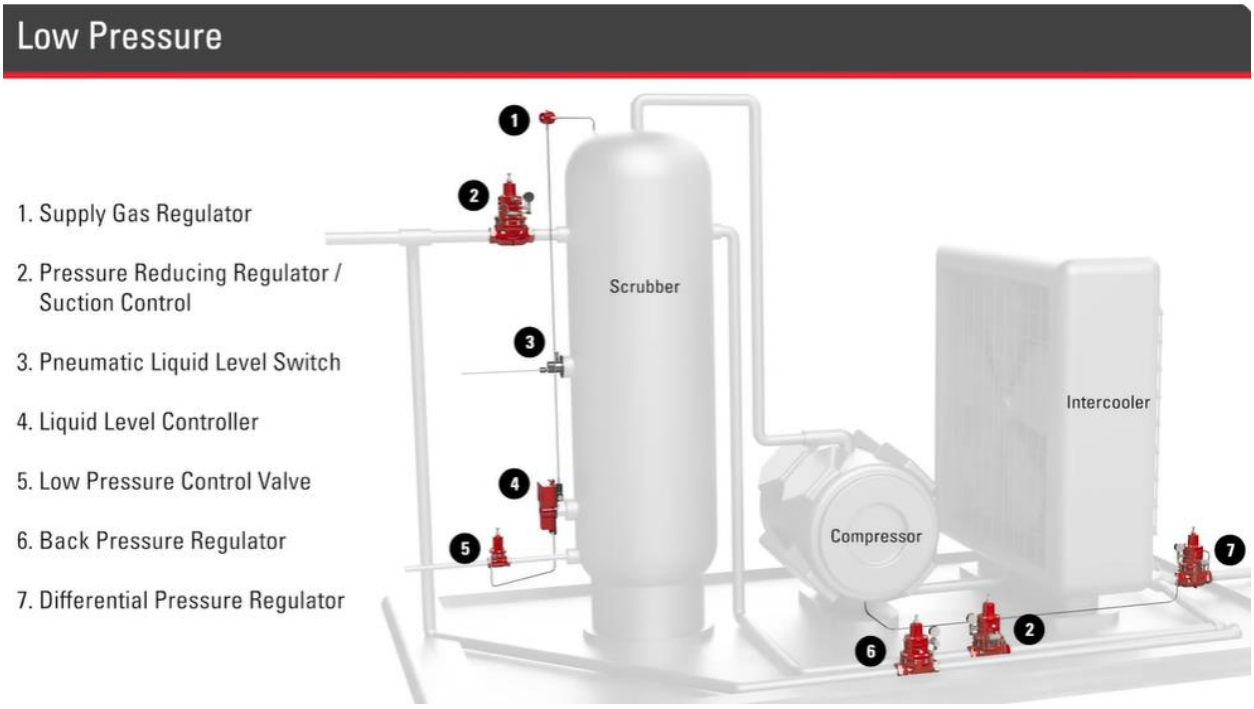
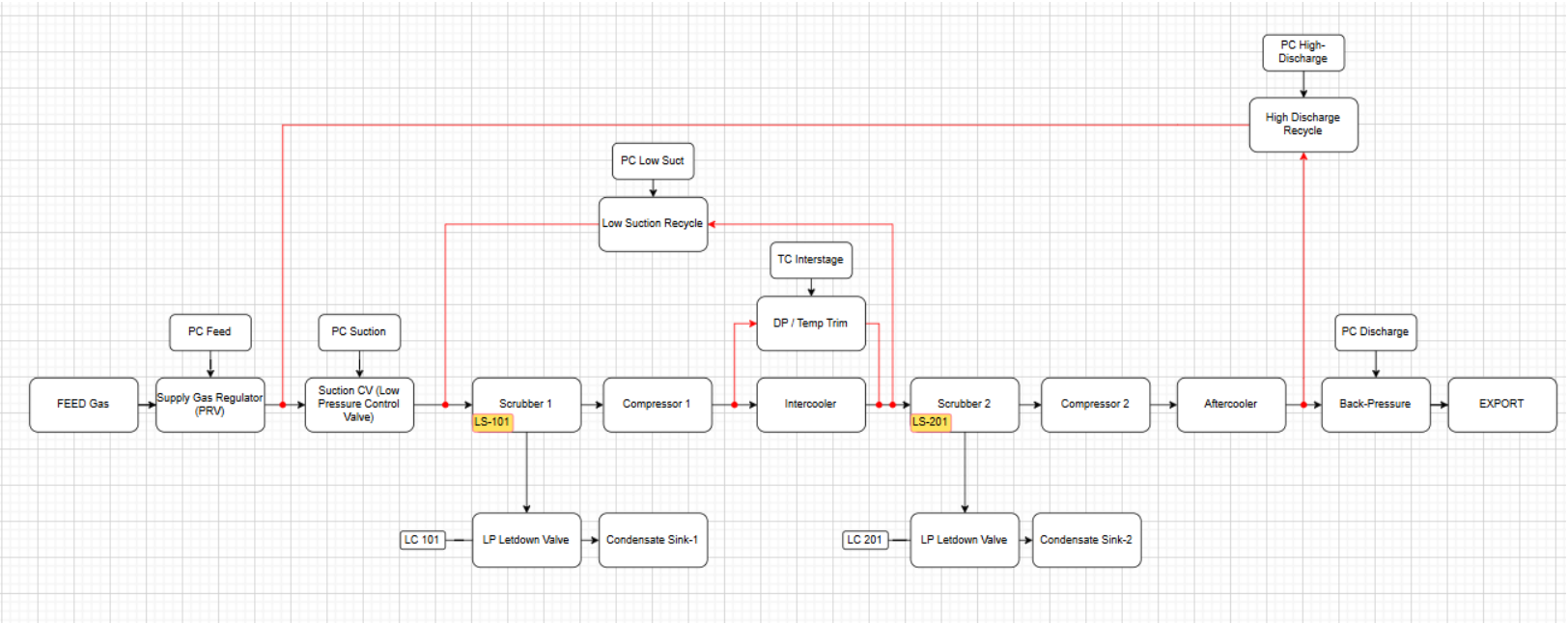
Simulator: Aspen Hysys

Compressor:





◇ Main Process Equipment

1. **Feed Gas**
    a. Source of natural gas entering the compression train.
2. **Scrubber-1 (Inlet KO Drum)**
    a. Removes liquids/condensate and solids before gas enters Compressor-1.
    b. Protects the compressor from liquid carryover.
3. **Compressor-1**
    a. First stage of compression. Increases pressure of the gas from scrubber-1.
4. **Intercooler**
    a. Cools the gas leaving Compressor-1 before it enters Scrubber-2.
    b. Reduces gas temperature, improves efficiency, and helps condense liquids.
5. **Scrubber-2 (Interstage KO Drum)**
    a. Separates condensed liquids formed after intercooling.
    b. Ensures Compressor-2 handles dry gas.
6. **Compressor-2**
    a. Second stage of compression. Brings the gas to final required discharge pressure.
7. **Aftercooler**
    a. Cools the gas after the second stage to reduce temperature before export.
8. **Export / Sales Gas Line**
    a. Final product gas stream leaving the system.

◇ Separation & Condensate Handling

9. **LP Letdown Valve (per scrubber)**
    a. Controls flow of separated liquids from each scrubber.
    b. Maintains scrubber liquid level within limits.
10. **Condensate Sink-1 / Sink-2**

- Collects separated liquid hydrocarbons or water from the scrubbers.

◇ Control Valves & Regulators

11. **Supply Gas Regulator (PRV)**

- Regulates inlet gas pressure.
- Prevents excess pressure from entering the compression train.
- Controlled by **PC-Feed**.

12. **Suction Control Valve (Low Pressure Control Valve)**

- Manages suction pressure entering Scrubber-1.
- Ensures stable compressor suction conditions.
- Controlled by **PC-Suction**.

13. **Back Pressure Valve**

- Holds a minimum discharge pressure before export.
- Protects compressors and ensures stable downstream conditions.
- Controlled by **PC-Discharge**.

◇ Recycle & Bypass Loops

14. **Low Suction Recycle Valve**

- Opens if suction pressure drops too low.
- Recycles gas from Intercooler outlet back to Scrubber-1 inlet.
- Prevents Compressor-1 surge.
- Controlled by **PC-LowSuct**.

15. **High Discharge Recirc Valve**

- Opens if discharge pressure gets too high.
- Recycles gas from Aftercooler outlet back upstream of the Suction CV.
- Prevents over-pressurization and surge.
- Controlled by **PC-HighDisch**.

16. **DP / Temp Trim Valve**

- Bypasses around the Intercooler.
- Regulates differential pressure or temperature between stages.
- Ensures stable operation of Scrubber-2 / Compressor-2 inlet.
- Controlled by **TC-Interstage**.

◇ Instrumentation & Controllers

17. **PC-Feed**
- Pressure controller for the Supply Gas Regulator.

18. **PC-Suction**
- Pressure controller for the Suction CV.

19. **PC-Discharge**
- Pressure controller for the Back-Pressure Valve.

20. **PC-LowSuct**
- Pressure controller for the Low Suction Recycle loop.

21. **PC-HighDisch**
- Pressure controller for the High Discharge Recirc loop.

22. **TC-Interstage**
- Temperature controller for the DP/Temp Trim loop around the Intercooler.

23. **LS-101 / LS-201 (Level Switches)**
- Provide alarms/trips if scrubber liquid level gets too high.

24. **LC-101 / LC-201 (Level Controllers)**
- Control the LP Letdown valves to keep scrubber liquid level stable.

SOP 01 - Day-to-Day Operations (Multi-Stage Centrifugal Compressor)

Who: Field/console operator (new hire friendly)
When: Each shift + on alarm
Goal: Keep the train stable, dry, cool, lubricated, and inside surge margins

A. Control room checks (at shift start, then hourly trending)

1. Review last shift log: trips/alarms/work permits.
2. DCS overview: confirm unit state = RUNNING, anti-surge loop in AUTO, setpoints per plan.
3. Record/Trend (log sheet or historian):
    - Suction **P/T/Flow**, Discharge **P/T**; ΔP = discharge–suction.
    - Recycle (anti-surge) valve % open; IGV angle or speed (RPM).
    - Lube oil **pressure & temperature**; bearing metal temps; **vibration** (X/Y on each bearing).
    - Suction KOD (knock-out drum) **level**. enggcyclopedia.com+1

B. Field walkdown (eyes/ears/nose; 5–10 min)

1. Listen/feel: no abnormal whine, rub, or rattling; no unusual vibration.
2. Look/smell: **no gas leaks**, no oil weeps around seals/console; area gas detectors normal.
3. Suction KOD: sight-glass/indicator shows low, steady level; auto drain functional (crack manual drain briefly if sticky). **Keep liquids out of suction.** Aqua Energy Expo+1
4. Aftercooler: fans turning / cooling water flowing; outlet temp within band.
5. Lube oil console: reservoir level OK; no high filter DP; no pump cavitation.
6. Seal gas panel (dry gas seal): supply pressure > process at seal by required margin; filters OK. CompressorTECH^2+1

C. During the shift (control in AUTO; gentle hands)

- Keep anti-surge margin healthy; avoid operating near the surge line (recycle will modulate).

- If throughput changes, adjust target **flow/pressure**-let capacity control (speed/IGVs) move first; recycle trims fine. Yokogawa
- Investigate and log any alarms (vibration, high temp, KOD level, oil pressure). Inst Tools

D. If something looks wrong (quick triage)

- **Rising vibration:** unload slightly (open recycle / reduce speed), watch trend; prepare for shutdown if it climbs.
- **High discharge T:** verify cooler/fans, unload; check for incipient surge.
- **Low lube-oil P:** ensure aux pump is on; if not recovering, **trip**. Never run without oil. ScienceDirect+1
- **Seal gas low:** restore supply/margin; if not immediate, **prepare to shut down**.

SOP 02 — Normal Startup & Normal Shutdown

A. Normal Startup (from depressured stop)

Pre-start safety & line-up

1. Work permits closed; area gas-free; ESDs tested "not in trip".
2. Suction/discharge **ESD closed**; blowdown/vent **closed**.
3. Drain liquids from suction KOD / intercooler low points; if opened to air, **nitrogen purge** suction per site procedure. Aqua Energy Expo

Support systems

4) Start **aux lube-oil pump**; verify pressure & temp within permissives; warm oil if required. ScienceDirect+1

5) Establish **seal-gas** (dry gas seal) margin; verify indicators. northerncal.swagelok.com

6) Start cooling (fans or CW).

Anti-surge & capacity set

7) Force **anti-surge valve (ASV) FULL OPEN**; set IGVs to start angle / speed at idle. Emerson

Spin & pressurize

8) Start driver (motor/turbine); monitor oil P, vibrations, temps.

9) Crack **suction ESD open** to admit gas; then crack **discharge** so check valve will open as head builds; keep ASV open to unload.

10) Ramp to operating speed/IGV; **close ASV gradually** as process flow is accepted; hand control back to AUTO. Yokogawa+1

Post-start

11) Verify all parameters steady; aux oil pump auto-standby; log startup data.

B. Normal Shutdown (planned)

1. Notify units upstream/downstream; reduce load target.
2. **Unload**: open ASV (increase recycle); reduce speed/IGV to minimum stable.
3. **Isolate**: close **suction** then **discharge** ESDs (no dead-head).
4. **Stop driver**; allow coast down with ASV open to avoid surge.
5. **Blowdown** the train to flare (auto or manual vent); verify pressure ≈ 0.

6. Keep **aux oil pump** running for cool-down, then stop; secure seal-gas per plan. Log stop.

SOP 03 — Manual Controls / Trip Response (Add-On Card)

If DCS/SCADA fails while running

- **Protect from surge:** Hand-wheel/bypass—OPEN recycle to maintain flow. Emerson Automation Experts
- **Capacity:** If VFD/IGV local panel exists, lower speed/close IGV slowly to control pressure.
- **Watch locals:** suction/discharge **PI/TI**, lube-oil **PI/TI**, KOD level, cooler outlet TI, bearing vibe meters. enggcyclopedia.com
- **If unstable:** perform **Normal Shutdown** steps manually; if not possible, hit **ESD**.

If automatic TRIP/ESD occurs

- Expect: driver stops, **inlet/outlet ESDs close, blowdown opens**; verify actions and keep aux oil pump on for coast down.
- Identify trip cause (vibe high, lube oil low, discharge P high, seal-gas low); do **not** restart until corrected. Inst Tools